

ROBINSON VIDVA, MBA, MS

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SUMMARY

Computational Biologist with 18+ years of experience in immunology, oncology, drug discovery, and data analysis. Expertise includes developing digital drug models, analyzing large-scale cancer genomics datasets, and leading cross-functional teams. Strong background in project management and data-driven decision making, with a focus on advancing innovation in computational biology and personalized medicine.

EDUCATION

M.S., Data Analytics, 2025 Western Governors University, USA

M.B.A., Project Management, 2020 Sikkim Manipal University, India

B.Tech., Biotechnology, 2007 Bharathidasan University, India

PROFESSIONAL EXPERIENCE

Research Technician Children's National Hospital, Washington, DC March 2025 - Present

Conducting translational research in pediatric neurodevelopment, integrating multi-omics approaches with computational methods to elucidate mechanisms underlying neonatal brain injury. Leading the development of automated behavioral phenotyping pipelines and applying machine learning frameworks to characterize motor and cognitive deficits in preterm birth models. Research contributions span RNA-seq, metabolomics, and genomics analyses linking molecular signatures to clinical outcomes in NEC, SIDS, and neurodevelopmental disorders.

Founder & Computational Biologist Digirobi Solutions, Bengaluru, India September 2019 - June 2022

Founded a computational biology consulting firm focused on translating complex biological data into actionable insights. Developed proprietary databases integrating cancer genomics, pharmacological, and nutritional data to support precision medicine applications. Engineered network models of SRC kinase signaling pathways and advanced cancer nutrigenomics research that culminated in a granted patent.

Senior Lead Scientist Cellworks Life, Bengaluru, India November 2017 - September 2019

Directed a multidisciplinary team of scientists in developing computational precision medicine platforms for oncology applications. Architected digital drug models spanning 10+ cancer types by integrating genomic, transcriptomic, and pharmacological data from TCGA, MSK, and proprietary sources. Established a comprehensive cancer cell line database comprising 500+ lines with drug sensitivity profiles, enabling systematic validation of computational predictions against clinical outcomes through peer-reviewed publications.

Lead Scientist Cellworks Life, Bengaluru, India January 2014 - October 2017

Spearheaded the development of an integrated drug knowledge repository encompassing target profiles, mechanisms of action, and pharmacokinetic parameters to accelerate therapeutic discovery. Translated preclinical findings into intellectual property, contributing to granted patents for PTEN-targeted cancer therapies. Evaluated therapeutic candidates for precision medicine applications while building strategic partnerships with pharmaceutical collaborators. Established mentorship programs and training curricula to advance team capabilities in computational biology.

Associate Lead Scientist Cellworks Life, Bengaluru, India June 2008 - December 2013

Pioneered the development of a systems biology model of Rheumatoid Arthritis incorporating 8 distinct cell types with comprehensive in-vivo validation. Identified synergistic drug combinations through systematic computational screening, advancing candidates to clinical collaboration. Research contributions resulted in 2 patents and presentations at the American College of Rheumatology, while collaborative studies on inflammatory skin disorders yielded peer-reviewed publications. Established foundational quality standards for computational model development and validation across the organization.

Associate Biomodeling Scientist Cellworks Life, Bengaluru, India July 2007 - May 2008

Constructed mechanistic models of immunological signaling in macrophages, incorporating 20+ interconnected pathways and gene regulatory networks. Developed rigorous validation frameworks through systematic perturbation analyses, establishing foundational methodologies for the organization's computational biology platform.

PUBLICATIONS

Peer-Reviewed Articles

1. Giannopoulos K, Karikis I, Byrd C, Sanidas G, Wolff N, Triantafyllou M, Simonti G, **Vidva R**, Koutroulis I, Theocharis S, Kratimenos P. Eph/Ephrin-Mediated Immune Modulation: A Potential Therapeutic Target for Skin Cancer. *Front Immunol.* 2025;16:1539567.
2. **Vidva R**, Raza MA, Prabhakaran J, Sheikh A, Sharp A, Ott H, Moore A, Fleisher C, Netherton H, Goldstein E, Pitychoutis PM. MyVivarium: A cloud-based lab animal colony management application with real-time ambient sensing. *Comput Struct Biotechnol J.* 2025;24:102-112.
3. Kratimenos P, Vij A, **Vidva R**, Koutroulis I, Delivoria-Papadopoulos M, Gallo V, Sathyanesan A. Computational analysis of cortical neuronal excitotoxicity in a large animal model of neonatal brain injury. *J Neurodev Disord.* 2022;14(1):19.
4. Fischer CL, Bates AM, Lanzel EA, Guthmiller JM, Johnson GK, Singh NK, Kumar A, **Vidva R**, Abbasi T, Vali S, Xie XJ, Zeng E, Brogden KA. Computational Models Accurately Predict Multi-Cell Biomarker Profiles in Inflammation and Cancer. *Sci Rep.* 2019;9(1):10877.
5. Drusbosky LM, **Vidva R**, Gera S, Lakshminarayana AV, Shyamasundar VP, Agrawal AK, Talawdekar A, Abbasi T, Vali S, Tognon CE, Kurtz SE, Tyner JW, McWeeney SK, Druker BJ, Cogle CR. Predicting response to BET inhibitors using computational modeling: A BEAT AML project study. *Leuk Res.* 2019;77:42-50.
6. Borgwardt DS, Martin AD, Van Hemert JR, Yang J, Fischer CL, Recker EN, Nair PR, **Vidva R**, Chandrashekariah S, Progulske-Fox A, Drake D, Cavanaugh JE, Vali S, Zhang Y, Brogden KA. Histatin 5 binds to Porphyromonas gingivalis hemagglutinin B (HagB) and alters HagB-induced chemokine responses. *Sci Rep.* 2014;4:3904.
7. Harvey LE, Kohlgraf KG, Mehalick LA, Raina M, Recker EN, Radhakrishnan S, Prasad SA, **Vidva R**, Progulske-Fox A, Cavanaugh JE, Vali S, Brogden KA. Defensin DEFB103 bidirectionally regulates chemokine and cytokine responses to a pro-inflammatory stimulus. *Sci Rep.* 2013;3:1232.

Preprints

1. Sanidas G, Simonti G, Ghaemmaghami J, Woyshner K, **Vidva R**, Wolff N, Triantafyllou M, Lowe C, Vij A, Pettersen HS, Chandereng T, Koutroulis I, Stein-O'Brien G, Sidiropoulos DN, Gallo V, Kratimenos P. Intrinsic gestational timing governs human cerebellar development after preterm birth. *bioRxiv.* 2025.

- Sanidas G, Simonti G, Ghaemmaghami J, Woyshner K, Wolff N, Byrd C, Triantafyllou M, Lowe C, Salisbury H, Goldstein E, Sathyanesan A, **Vidva R**, Koutroulis I, Stein-O'Brien G, Sidiropoulos DN, Gallo V, Kratimenos P. Prematurity Reprograms Cerebellar Development and Long-Term Behavior. *bioRxiv*. 2025.

PATENTS

Granted

- Vali S, Abbasi T, Stopka T, Minarik L, Singh NK, Usmani S, Radhakrishnan S, Sikora H, **Vidva R**, Pimkova K. Use of scientifically matched plant supplements combined with antineoplastic compounds for treatment of hematological malignancies. US11478444B2. 2022.
- Vali S, Usmani S, Sultana Z, Kumar A, Abbasi T, **Vidva R**. Combination of nelfinavir, metformin and rosuvastatin for treating cancer caused by aberrations in PTEN/TP53. US10098880B2. 2018.

Applications

- Vali S, Abbasi T, Fernandes P, Rajagopalan S, Alam A, **Vidva R**, et al. System and method for development of therapeutic solutions. US20150302167A1. 2015.
- Vali S, **Vidva R**, Nair PR, Fernandes P, Abbasi T, Radhakrishnan S. Compositions, process of preparation of said compositions and method of treating inflammatory diseases. US20150098993A1. 2015.
- Vali S, **Vidva R**, Nair PR, Fernandes P, Abbasi T, Radhakrishnan S. Compositions, process of preparation of said compositions and method of treating inflammatory diseases. US20140127295A1. 2014.

CONFERENCE PRESENTATIONS

- Sanidas G, Lowe C, Byrd C, **Vidva R**, Simonti G, Chandereng T, Gallo V, Kratimenos P. Necrotizing Enterocolitis Selectively Disrupts Pontine Development in the Preterm Brain. Society for Neuroscience (SfN), 2025. Poster.
- Simonti G, **Vidva R**, Sanidas G, Byrd C, Lowe C, Triantafyllou M, Wolff N, Chandereng T, Koutroulis I, Gallo V, Kratimenos P. Early postnatal rehabilitation mitigates motor, growth, and social deficits in perinatal brain injury. Society for Neuroscience (SfN), 2025. Poster.
- Cogle CR, Abbasi T, Singh NK, Sauban M, Raman RK, **Vidva R**, et al. AraC-Daunorubicin-Etoposide (ADE) Response Prediction in Pediatric AML Patients. American Society of Hematology (ASH), 2018.
- Tyner JW, Druker BJ, Tognon CE, Kurtz SE, Drusbosky LM, Sahu D, **Vidva R**, et al. Predicting Response to Dasatinib Using a Computational Model: A Beat AML Project Study. American Society of Hematology (ASH), 2018.
- Tyner JW, Druker BJ, Kurtz SE, Tognon CE, Drusbosky LM, **Vidva R**, et al. Predicting Response to BET Inhibitor in Combination with Palbociclib/Sorafenib. American Society of Hematology (ASH), 2018.
- Drusbosky LM, Turcotte M, Castillo P, **Vidva R**, et al. Predicting Response to CDK4/6 Inhibitors Using a Computational Biology Model. American Society of Hematology (ASH), 2017.
- Drusbosky LM, **Vidva R**, et al. Predicting Response to BET Inhibitors Using a Computational Model: A Beat AML Project Study. American Society of Hematology (ASH), 2017.

8. Drusbosky LM, **Vidva R**, et al. Predicting Response to IDH1/IDH2 Inhibitors Beyond IDH Mutations. American Society of Hematology (ASH), 2017.
9. Drusbosky L, Abbasi T, Vali S, Radhakrishnan S, Kumar Singh N, Usmani S, Parashar D, **Vidva R**, et al. A Genomic Signature Predicting Venetoclax Treatment Response in AML. American Society of Hematology (ASH), 2016.
10. Fischer C, Abbasi T, Vali S, Nair P, **Vidva R**, Tiwari K, Brogden KA. A Predictive Model of an Oral Inflammatory Response. AADR/CADR, 2014.
11. Harvey L, Recker E, Raina M, Radhakrishnan S, Prasad S, **Vidva R**, Cavanaugh J, Vali S, Brogden KA. HBD3 Both Enhances, Attenuates Dendritic Cell MAPK Responses to HagB. IADR/AADR/CADR, 2013.
12. Singh G, **Vidva R**, Nair P, Radhakrishnan S, Fernandes P, Abbasi T, Refino C, Cruz JD, Vali S. Predictive Software-Based Mathematical Modeling: A Novel Approach to Development of Oral Therapies for Rheumatoid Arthritis. ACR/ARHP, 2012.
13. Vali S, Hegedus C, Bhattacharjee C, **Vidva R**, Schmelzer K, Inceoglu B, Hammock BD. Insights into the Anti-inflammatory Action of the Soluble Epoxide Hydrolase Inhibitor through Systems Biology. FASEB, 2008.

PROFESSIONAL MEMBERSHIPS

International Society for Computational Biology (ISCB), 2024 - Present

TECHNICAL SKILLS

Programming: Python (matplotlib, seaborn, pandas, scikit-learn), R (ggplot2, DESeq2, Seurat), SQL, MATLAB, C/C++, Unix Shell

Bioinformatics: Multi-omics analysis, pathfindR, MetaboAnalyst, CellPhoneDB, spatial transcriptomics (Merscope, Xenium)

Behavioral Analysis: DeepLabCut, BSOD, BONSAI, Erasmus Ladder

Data Science: Machine learning, predictive modeling, clustering, trajectory analysis, mixed-effects models, time-series, survival analysis, Tableau

Software Development: Web application development, database design, cloud-based systems, IoT integration

CERTIFICATIONS

Specializations: Cancer Biology (Johns Hopkins University), Drug Development Product Management (UC San Diego), Biomedical Big Data (Georgetown University)

Professional Certificates: Advanced Data Modeling, Data Analytics (Western Governors University)

LANGUAGES

English (Fluent), Tamil (Native), Kannada (Conversational)

REFERENCES

Available upon request